

Functional Requirements Document (FRD)

Project Title

Employee Attrition Analysis & Retention Intelligence System

1. Purpose

The purpose of this document is to define the functional and non-functional requirements of the Employee Attrition Analysis & Retention Intelligence System.

The system is designed to help organizations understand employee attrition patterns, identify the factors influencing employee turnover, and support data-driven retention strategies through analytics, reporting, and predictive insights.

This document serves as a bridge between business requirements and system implementation.

2. System Overview

The Employee Attrition Analysis & Retention Intelligence System will analyze employee data and provide meaningful insights into workforce behavior.

The system will help stakeholders:

- Monitor employee attrition trends
- Identify departments with high turnover
- Analyze factors contributing to attrition
- Predict employees at risk of leaving
- Support retention planning and workforce decision-making

3. User Roles

User Role	Responsibility
HR Manager	Monitor attrition trends and employee retention
Department Manager	Review department-level attrition insights

Executive Management	Make strategic workforce decisions
Business Analyst	Analyze data and generate reports
HR Executive	Support retention initiatives

4. Functional Requirements

FR-001: Employee Data Analysis

Description

The system shall analyze employee records available within the dataset.

Priority

High

Business Value

Provides the foundation for all workforce analytics.

FR-002: Overall Attrition Rate Calculation

Description

The system shall calculate the overall employee attrition rate.

Priority

High

Business Value

Helps management understand the scale of employee turnover.

FR-003: Department-Wise Attrition Analysis

Description

The system shall display attrition statistics for each department.

Priority

High

Business Value

Identifies departments requiring immediate attention.

FR-004: Job Role Attrition Analysis**Description**

The system shall display attrition trends across different job roles.

Priority

High

Business Value

Supports workforce planning and role-specific retention strategies.

FR-005: Salary Analysis**Description**

The system shall analyze employee compensation and its relationship with attrition.

Priority

High

Business Value

Helps identify compensation-related retention issues.

FR-006: Overtime Analysis**Description**

The system shall analyze the impact of overtime on employee attrition.

Priority

High

Business Value

Supports employee well-being and workload management initiatives.

FR-007: Demographic Analysis

Description

The system shall analyze attrition based on age, gender, education, and marital status.

Priority

Medium

Business Value

Provides demographic insights into workforce behavior.

FR-008: Employee Experience Analysis

Description

The system shall analyze employee tenure, total work experience, and promotion history.

Priority

High

Business Value

Helps understand the relationship between career progression and attrition.

FR-009: Dashboard Visualization

Description

The system shall provide interactive dashboards for data visualization.

Priority

High

Business Value

Supports faster and more effective decision-making.

FR-010: Data Filtering Capability

Description

The system shall allow filtering based on:

- Department
- Job Role
- Gender
- Age Group
- Attrition Status
- Overtime Status

Priority

Medium

Business Value

Allows detailed analysis for specific employee groups.

FR-011: Business Insights Reporting

Description

The system shall generate reports and business insights from analyzed data.

Priority

High

Business Value

Supports management reporting and strategic planning.

FR-012: Employee Risk Identification

Description

The system shall identify employees who are likely to leave the organization.

Priority

High

Business Value

Enables proactive retention initiatives.

FR-013: Attrition Prediction

Description

The system shall use historical employee data to predict future attrition risk.

Priority

High

Business Value

Supports workforce planning and future resource allocation.

FR-014: Retention Recommendations

Description

The system shall provide recommendations based on identified attrition drivers.

Priority

Medium

Business Value

Supports implementation of effective retention strategies.

5. Dashboard Requirements

Executive Dashboard

The dashboard shall display:

- Total Employees
- Total Attrition
- Attrition Rate
- Average Monthly Income
- Average Years at Company

Department Dashboard

The dashboard shall display:

- Attrition by Department
- Attrition by Job Role

- Department Performance Overview

Employee Dashboard

The dashboard shall display:

- Attrition by Age Group
- Attrition by Gender
- Attrition by Marital Status
- Attrition by Education Field

Retention Dashboard

The dashboard shall display:

- Overtime Impact Analysis
- Job Satisfaction Analysis
- Work-Life Balance Analysis
- Promotion Analysis

Predictive Dashboard

The dashboard shall display:

- Employee Risk Scores
- Attrition Prediction Results
- High-Risk Employee Segments

6. Acceptance Criteria

FR-002: Overall Attrition Rate

Acceptance Criteria

- Attrition percentage must be displayed.
- Calculation should be based on available employee records.
- Dashboard should refresh automatically when data is updated.

FR-003: Department Analysis

Acceptance Criteria

- All departments must be displayed.
- Attrition count and percentage must be visible.
- Users should be able to compare departments.

FR-009: Dashboard Visualization

Acceptance Criteria

- Dashboard must be interactive.
- Filters should work correctly.
- Visualizations should update dynamically.

FR-013: Attrition Prediction

Acceptance Criteria

- System must generate attrition predictions.
- Risk categories should be displayed.
- Prediction results should be visible in reports.

7. Non-Functional Requirements

ID	Requirement
NFR-01	Dashboard should load within 5 seconds
NFR-02	Data accuracy should be maintained
NFR-03	Reports should be easy to understand
NFR-04	System should support future dataset expansion
NFR-05	Dashboard should be user-friendly

8. Assumptions

- Employee data is accurate and complete.
- Historical records are available for analysis.
- Stakeholders will use generated reports for decision-making.
- Business recommendations will be based on available data.

9. Success Metrics

The project will be considered successful if:

- Key attrition drivers are identified.
- High-risk employee groups are recognized.
- Dashboards provide actionable insights.
- Predictive analysis supports workforce planning.
- Retention recommendations are generated.

10. Expected Deliverables

- Project Charter
- Business Questions Document
- BRD
- FRD
- User Stories
- Process Flow Diagrams
- SQL Analysis
- Power BI Dashboard
- Attrition Prediction Model
- Business Insights Report
- Final Presentation

Document Information

Document Type: Functional Requirements Document